



School of Engineering and Engineering Technology
Department of Information Communication Technology
Second Semester Examination 2023/2024 Session
ICT 502 Telecommunication Software Development

Instruction: Answer only Four (4) Questions

Unit 3.0

Time: 180 minutes.

Question One

- (a) Explain what software development in telecommunications entails, including the stages of the Software Development Life Cycle Project (SDLCP) and their significance. Provide relevant examples of telecommunication software and how these stages are applied. (20 marks)
- (b) Draw a detailed diagram of a typical software development process. Label each stage. (5marks)

Question Two

- (a) Explain the software development maturity model, including its purposes and benefits. (10 marks)
- (b) Discuss the importance of notification systems in telecommunication software development, including types of notifications and the role of notification models. (8marks)
- (c) As a telecommunication software developer, explain the significance of Model-Driven Development (MDD) in telecommunication software development. Discuss how it contributes to the overall development process. (7marks)

Question Three

- (a) Explain how market trends, user behavior, and technological forecasts guide strategy and decision-making in telecom software development. Provide relevant industry examples to illustrate your points. (10marks)
- (b) What are the key stages in crafting a telecom strategy? Summarize situation analysis, goal setting, strategy formulation, and implementation. Illustrate with industry examples. (7marks)
- (c) Discuss the main challenges in predictions and strategy definition for telecom software development. Include examples of mitigation strategies and best practices to address these challenges. (8marks)

Question Four

- (a) Identify key areas where research and development (R&D) has contributed to the growth of Nigeria's telecommunication sector. (7 marks)

*funding & investment
talent & skills
infrastructure and resources*

(b) Analyze obstacles hindering ICT R&D progress in Nigeria and suggest strategies to overcome them. (9 marks)

(c) Discuss how Nigerian R&D can influence or contribute to global ICT trends. Additionally, identify specific areas where international R&D developments could be tailored to meet Nigeria's unique needs. (9 marks)

Question Five

(a) What is the role of intellectual property (IP) in telecom software development? Discuss how IP protects innovations and contributes to competitive advantage, including examples from Nigeria and globally. (10 marks)

(b) Describe the process and best practices for patent search and clearance in telecom software development. How do these practices help avoid infringement and ensure the freedom to operate? (8 marks)

(c) Discuss the challenges of managing intellectual property (IP) in telecom software development. How can companies address these challenges, and what are some effective strategies and best practices? (7 marks)

Question Six

(a) Explain the role of research in analyzing business ideas, and provide examples from the telecom industry in Nigeria. (5 marks)

(b) Discuss the types of research and how each type is used in evaluating business ideas. Provide examples where necessary. (5 marks)

(c) A telecom software company invests ₦15 million in a mobile payment platform. They project that within a year, the platform will generate ₦22 million in revenue.

(i) Calculate the Return on Investment (ROI) for the project. (9 marks)

(ii) Discuss the implications of the ROI value obtained for investors and shareholders, and how you would motivate more investors based on this ROI. (6 marks)

*market
technical
economic*

*identify problem & conditions
conduct market research
technical feasibility
economic feasibility*



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SCHOOL OF ELECTRICAL SYSTEMS ENGINEERING
DEPARTMENT OF INFORMATION & COMMUNICATION ENGINEERING
B. ENG. DEGREE TEST SECOND SEMESTER 2024/2025 SESSION

COURSE CODE: ICT 502 COURSE TITLE: Telecommunication Software Development

UNITS: 3 TIME ALLOWED: 60 minutes

Instruction: Answer ALL Questions (20 Marks)

Question One (10 marks)

An ICT solutions company is partnering with FUTA to pilot a cloud-based e-learning platform for digital education across the university campus. The project requires an initial capital investment of ₦30 million. It is projected to generate ₦18 million annual revenue for the first three years, with an annual operating cost of ₦5 million. The company's cost of capital is 12% per annum.

(a) Calculate:

i. Annual net cash inflow.

ii. Net Present Value (NPV) over the three years. $\rightarrow 18$

iii. Interpret your results in terms of investment viability. 13

$18 -$ (4 marks)

$= -0.4$ (4 marks)

$= 0.255$ (2 marks)

Question Two (10 marks)

(a) Describe the main phases of Rapid Application Development (RAD) with suitable Nigerian industry examples. $\begin{matrix} \text{Initial} \\ \Sigma \\ \text{Question} \\ \text{Re} \end{matrix}$ (5 marks)

(b) Define Intellectual Property (IP) and explain its importance in protecting telecom software innovations, giving one Nigerian and one global example. $\begin{matrix} \text{Competition} \\ \text{revenue} \end{matrix}$ (5 marks)



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B. ENG. DEGREE EXAMINATIONS. SECOND SEMESTER 2024/2025 SESSION

COURSE CODE: ICT 502 **COURSE TITLE:** Telecommunication Software Development
UNITS: 3

TIME ALLOWED: 180 minutes

Instruction: Answer Only Four (4) Questions

Question One

A Nigerian Telecom Company is partnering with FUTA to pilot a 5G-enabled IoT platform for smart-city applications within the university campus. The project requires an initial capital investment of ₦48 million. It is projected to generate ₦65 million annual revenue for the first three years, with an annual operating cost of ₦12 million. The company's cost of capital is 10% per annum.

(a) Calculate:

i. Annual net cash inflow.

$$= \text{Revenue} - \text{Operating Cost} = 65 - 12 = 53$$

0.7

4 marks

ii. Net Present Value (NPV) over three years.

$$= \frac{53}{(1+0.1)^1} + \frac{53}{(1+0.1)^2} + \frac{53}{(1+0.1)^3} - 48 = 10.729$$

0.4

6 marks

iii. Payback period.

0.729

5 marks

iv. Return on Investment (ROI) over the project's three-year lifespan

$$= \frac{10.729}{48} = 0.2235$$

5 marks

(b) Interpret the results in terms of investment viability and discuss how such a pilot could support FUTA's role as a technology innovation hub in Nigeria's smart-city agenda.

5 marks

Question Two

(a) Discuss the role of cloud computing in telecommunication software development. Highlight its impact on Nigerian operators (scalability, cost efficiency, service delivery) with examples.

15 marks

(b) Identify and explain three key security challenges in cloud-based telecom solutions, and suggest mitigation strategies suitable for Nigerian telecom companies.

10 marks

Question Three

(a) Explain what telecommunication software development entails. Include its main application areas and relevant industry examples, with at least one example relevant to Nigeria or FUTA's ICT/ICE department.

15marks

(b) Using the Software Development Life Cycle Project (SDLCP), describe each stage briefly and highlight its significance in telecommunication software development. Provide one example of how a stage could be applied in a FUTA-based ICT research project.

10 marks

① Requirement gathering

② Analysis, functional decomposition

③ Design and Architecture development

④ Implementation and coding

⑤ Software testing

⑥ Software release deployment

⑦ Software maintenance and documentation

⑧ Evaluation and retirement



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Question Four

- (a) Describe the main phases of Rapid Application Development (RAD) with suitable Nigerian industry examples. **12 marks**
- (b) Identify and explain any two benefits and two limitations of RAD in telecom software projects. Relate one limitation to a possible challenge in a FUTA student-driven project. **8 marks**
- (c) Briefly outline the four types of software maintenance with Nigerian examples. **5 marks**

Question Five

- (a) Explain the concept and purpose of Software Development Maturity Models. Illustrate with examples from CMMI, including at least one application relevant to Nigeria's telecom sector. **12 marks**
- (b) Discuss the role of notification systems in telecommunication software development. Give examples from Nigeria (e.g., MTN, JAIZ Bank, GTBank) and relate how such a system could benefit FUTA's internal communications or student services. **8 marks**
- (c) Define Model-Driven Development (MDD) and explain two benefits it offers in telecommunication software development. Provide an example of how FUTA's ICT/ICE students could apply MDD in a capstone project. **5 marks**

Question Six

- (a) Define intellectual property (IP) and explain its importance in protecting innovations in telecom software development. Relate your answer to academic research outputs, novel designs, and prototypes developed within universities such as FUTA. Provide one example from Nigeria and one from the global industry. **10 marks**
- (b) Describe the process of patent search and clearance in telecom software development. Explain how this process can be applied in an academic context to protect university research innovations and student-led projects, using at least one example from Nigeria. **8 marks**
- (c) Identify two challenges in managing IP arising from academic research in Nigerian institutions and propose practical solutions for safeguarding and commercialising such innovations. **7 marks**